Geometry-Conditioned Site Refinement Specification

Stage 11E5b

mdstats

2026-07-25

Contents

Scope

Stage 11E5b is an optional refinement downstream of the frozen Stage-11E5 statistical catalog and Stage-11E5a structural fingerprints. It asks whether a validated site's center follows independently measured framework geometry. The implementation owner is:

```
mdstats.analysis.density.geometry_conditioning
```

The stage consumes:

```
ValidatedFrozenCatalog              Stage 11E5 state identity and block plan
CoordinationFingerprintCatalog      Stage 11E5a exact local ion coordinates
FrameworkPredictorTable             framework-only structural descriptors
FrozenRegionDefinition              frozen center, core, and basin radii
```

It produces one `GeometryConditionedSiteCatalog`. The dynamic model may move a site center and rigidly translate its nested core and basin, but it never changes the persistent statistical-state identity, redefines the framework frame, or iteratively reassigns discovery samples.

This stage does **not** publish final residence or transition events, rates, barriers, a PMF, or a kinetic network. Those remain Stage 11E6 and later work.

Borrowed methods and package-specific constructions

Weighted affine regression and held-out model comparison are standard statistical methods. The regression and bias–variance background follows Hastie, Tibshirani, and Friedman, *The Elements of Statistical Learning*, second edition (2009). The separation of model fitting and independent predictive assessment is standard cross-validation/model-selection practice; see Stone (1974), *Journal of the Royal Statistical Society B* **36**, 111–147.

The following are mdstats-specific constructions:

- exact E5/E5a/registered-structural-view source binding;
- a framework-only predictor contract that rejects mobile-ion-derived features;
- one-pass fitting on frozen discovery assignments;
- selection-block comparison followed by untouched final-validation confirmation;

- rigid translation of a frozen nested region rather than implicit use of the frozen basin around a moving center;
- simultaneous static and dynamic membership records;
- explicit comoving, center, and boundary displacement diagnostics;
- an exclusive assignment-conflict vocabulary; and
- lower/upper occupancy bounds under moving-region overlap.

Source and predictor contracts

For state i and structural association a, the fingerprint, predictor table, and frozen region must share:

```
state_id
candidate_index
persistent_identity
sample_indices
frame_indices
registered_structural_view_digest
validated_frozen_catalog_signature
```

Only one selected structural association per statistical state enters one E5b catalog. Alternative E5a associations remain available in the E5a catalog and may be evaluated in separate candidate runs; they are not silently merged.

A `FrameworkPredictorTable` stores one row per E5a sample and one column per structural feature. Examples include:

- ring aperture, area, perimeter, and puckering;
- oxygen-class radii;
- rank-safe structural harmonic coefficients;
- gauge-defined structural phase coordinates when supported;
- tile or cage volume; and
- local framework strain descriptors.

The table carries `framework_only=true`. Mobile-ion positions, ion-derived coordination distances, provisional site labels, or residuals computed from the same mobile-ion target are forbidden as predictors. This prevents the structural frame from being redefined by the ion whose position is being predicted.

Static and affine center models

Let $\mathbf{y}_n \in \mathbb{R}^3$ be the ion coordinate in the persistent local structural frame and $\boldsymbol{\xi}_n \in \mathbb{R}^p$ the framework descriptor. The frozen static model is

$$\hat{\mathbf{c}}_n^{\mathrm{static}} = \mathbf{c}_0.$$

The candidate framework-conditioned model is

$$\hat{\mathbf{c}}_n^{\mathrm{dyn}} = \mathbf{b} + B\left(\boldsymbol{\xi}_n - \overline{\boldsymbol{\xi}}_D\right),$$

where D is the discovery block. With represented-time weights w_n, E5b solves

$$\min_{\mathbf{b},B} \sum_{n\in D} w_n \parallel \mathbf{y}_n - \mathbf{b} - B(\boldsymbol{\xi}_n - \overline{\boldsymbol{\xi}}_D) \parallel^2 .$$

The fitted record stores:

```
predictor mean
intercept
coefficient matrix
fit rank and parameter count
condition number
sample count
weighted residual RMS
residual covariance
```

The fit fails closed when discovery support is insufficient, the weighted design is rank deficient, or its condition number exceeds the declared threshold. Regularization is explicit and defaults to zero; when enabled, the intercept is not penalized.

Noncircular selection and validation

The E5 block plan supplies disjoint or explicitly provenance-labeled frame sets:

```
discovery
selection
final_validation
```

Discovery assignments are the frozen E5/E5a samples. They are not recomputed after fitting. For block R, define

$$\mathrm{RMS}_R(m) = \left[\frac{\sum_{n\in R} w_n \parallel \mathbf{y}_n - \hat{\mathbf{c}}_{m,n} \parallel^2}{\sum_{n\in R} w_n}\right]^{1/2}.$$

The selection improvement is

$$I_S = \frac{\mathrm{RMS}_S(\text{static}) - \mathrm{RMS}_S(\text{dynamic})}{\max[\mathrm{RMS}_S(\text{static}), \epsilon]}.$$

The dynamic model is eligible only when I_S exceeds the declared minimum. It is retained only when the untouched validation block has adequate support and

$$\mathrm{RMS}_V(\text{dynamic}) \le (1 + \delta_V)\mathrm{RMS}_V(\text{static}),$$

where δ_V is the maximum admitted validation degradation. Otherwise the static model remains selected and the exact reason is stored:

```
static_retained
dynamic_retained
insufficient_discovery_support
insufficient_selection_support
independent_validation_unavailable
final_validation_contradicted
rank_deficient
ill_conditioned
unresolved
```

Selection success is therefore not itself final confirmation.

Moving nested regions

A `FrozenRegionDefinition` contains a center $\mathbf{c}_0$, core radius r_C, and basin radius r_B satisfying

$$0 < r_C < r_B.$$

The first implementation translates the frozen shape without changing its radii or covariance:

$$C_i(t) = \{\mathbf{y} : \parallel \mathbf{y} - \hat{\mathbf{c}}_i(t) \parallel \le r_C\},$$

$$B_i(t) = \{\mathbf{y} : \parallel \mathbf{y} - \hat{\mathbf{c}}_i(t) \parallel \le r_B\}.$$

A moving center therefore never uses a basin still centered at $\mathbf{c}_0$. Shape-conditioned or covariance-conditioned regions are explicitly deferred.

Every sample retains three memberships:

```
static_membership
dynamic_membership
selected_membership
```

The selected membership uses the dynamic region only when the dynamic model passes the selection and final-validation gate. Static and candidate-dynamic memberships remain available as counterfactual diagnostics.

Motion and crossing diagnostics

For adjacent samples of the same atom and source segment, E5b records

$$\Delta \mathbf{y}_n = \mathbf{y}_n - \mathbf{y}_{n-1},$$

$$\Delta \mathbf{c}_n = \hat{\mathbf{c}}_n^{\mathrm{dyn}} - \hat{\mathbf{c}}_{n-1}^{\mathrm{dyn}},$$

and the local comoving displacement

$$\Delta \mathbf{y}_n^{\mathrm{comov}} = (\mathbf{y}_n - \hat{\mathbf{c}}_n^{\mathrm{dyn}}) - (\mathbf{y}_{n-1} - \hat{\mathbf{c}}_{n-1}^{\mathrm{dyn}}).$$

For a rigidly translated spherical region, the boundary displacement magnitude is $\| \Delta \mathbf{c}_n \|$. A dynamic membership change is classified as `boundary_induced` only when the frozen membership does not change and the ion movement is small relative to the boundary motion. Other changes remain `ion_driven`, `mixed`, or `unresolved`. A boundary-swept crossing is never silently counted as a purely ion-driven jump.

Crossing diagnostics never bridge atom identities or E0b source-segment resets.

Assignment conflicts and occupancy bounds

Moving regions belonging to distinct states may overlap. Each represented sample receives one exclusive `AssignmentConflictStatus`:

```
unique_core
unique_basin
multiple_core_overlap
multiple_basin_overlap
static_dynamic_conflict
outside_supported_regions
assignment_unresolved
```

No ion is double-counted in the lower occupancy estimate. For state i,

$$W_i^{\mathrm{lower}} = \sum_n w_n \mathbf{1}\{n \text{ is uniquely assigned to } i\},$$

$$W_i^{\mathrm{upper}} = \sum_n w_n \mathbf{1}\{n \text{ belongs to any selected region of } i\}.$$

The result stores lower and upper fractions, core-overlap fraction, basin-overlap fraction, and unresolved fraction. Declared overlap gates are reported in catalog diagnostics; they do not silently delete a state.

Residual covariance

E5b reports weighted residual covariance before and after conditioning:

$$\Sigma_m = \frac{1}{\sum_n w_n} \sum_n w_n (\mathbf{r}_{m,n} - \bar{\mathbf{r}}_m)(\mathbf{r}_{m,n} - \bar{\mathbf{r}}_m)^T.$$

A reduced trace is supporting evidence for framework-conditioned motion, not a standalone validation decision. The held-out residual gate remains authoritative.

Persistence, resources, and serialization

All option, predictor, region, model, score, crossing, state, conflict, occupancy, and catalog records are immutable and SHA-256 signed. Deserialization recomputes every nested signature and rejects tampering.

Resource preflight bounds:

```
states
samples
predictor values
crossing records
conflict records
serialized records
```

The preflight occurs before aggregate conflict and occupancy construction.

Acceptance tests

The focused acceptance suite must demonstrate:

1. a true framework-following center is selected and confirmed on held-out data;
2. a selection gain contradicted by final validation leaves the static model selected;
3. rank-deficient predictors fail closed;
4. mobile-ion-dependent predictors are rejected;
5. static and dynamic memberships remain jointly available;
6. a boundary-swept crossing is not labeled ion driven;
7. overlap produces exclusive conflicts and occupancy bounds;
8. source mismatch, resources, serialization, and tamper checks fail closed; and
9. public package exports remain stable.

Deferred work

Stage 11E6 owns final core–basin hysteresis, residence intervals, transition intervals, recrossing policy, censoring, and final event statistics. E5b records candidate memberships and motion diagnostics only.